



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Mapping the evidence with R

R practical — heat maps and cross-
tabulations for systematic maps

Damien Beillouin
CIRAD, UPR HortSys

SOUTIENS INSTITUTIONNELS





Learning objectives

By the end of this practical, you will be able to:

- 1 **Count studies, not rows**, when describing an evidence base.
- 2 Build an **evidence heat map** that shows clusters **and gaps**.
- 3 **Cross a third variable** from your codebook (organisms, validity...).
- 4 Say what an evidence map **can and cannot** tell you.

From codebook to map

This morning you designed a **codebook**. Each coded variable can become an **axis** of a map.

Today's data: a published systematic map on **diversified farming and biodiversity** (Jones et al. 2021).

- **237** studies · **4,076** comparisons
- **48** countries · 1986–2021
- 105 coded columns in the full database (32 kept for today): practice, comparator, taxon, functional group, validity...

Our question

Where is the evidence – and where is it missing – on the effects of farm diversification on biodiversity?

Coding shapes the map

The published dictionary lists “cover crop”; the data contain none. The data contain “Associated plant species”; the dictionary does not define it.

Quick quiz

SHOW OF HANDS

The dataset has 360 rows from 1992. How many studies is that?

360

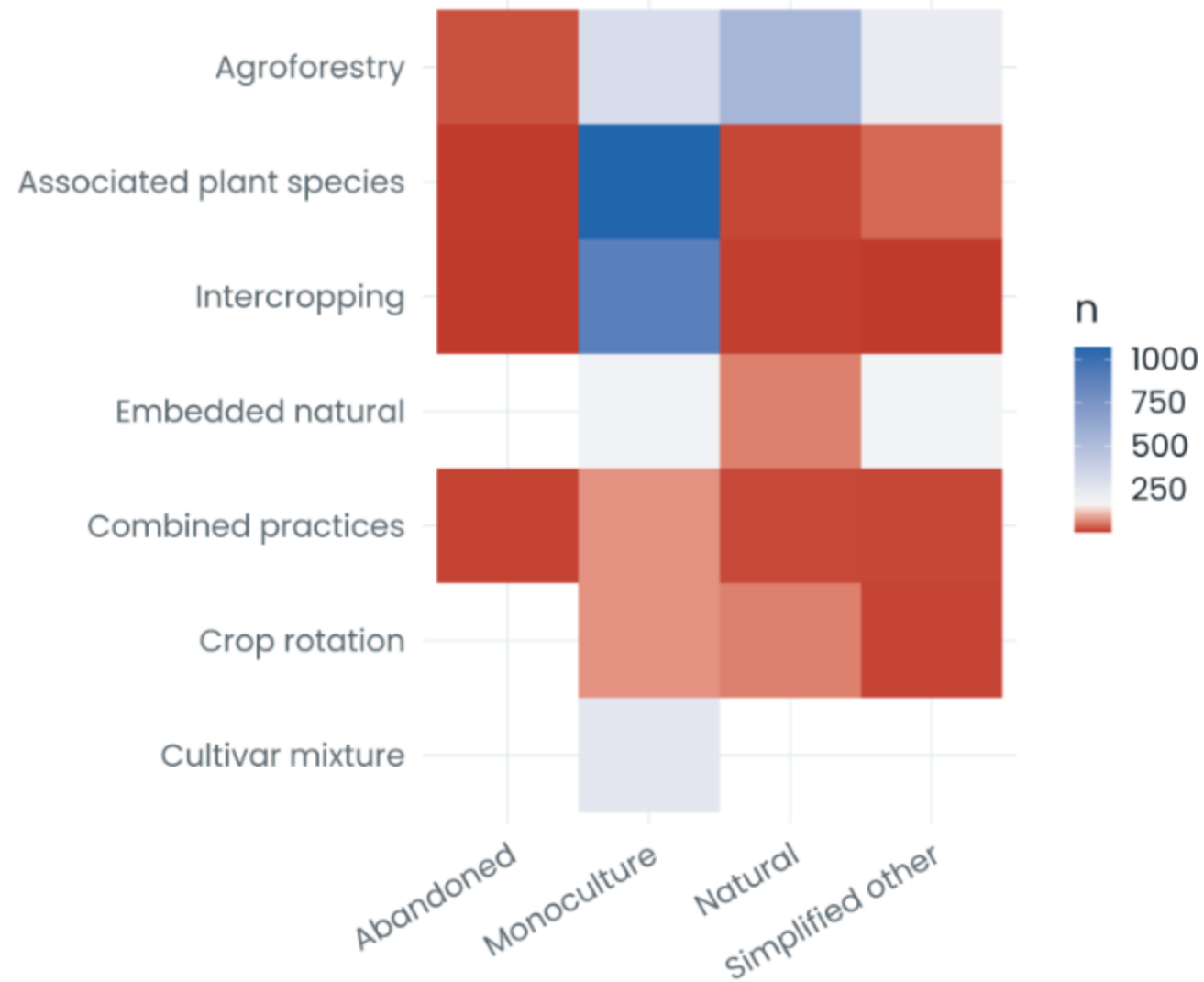
About 36

2

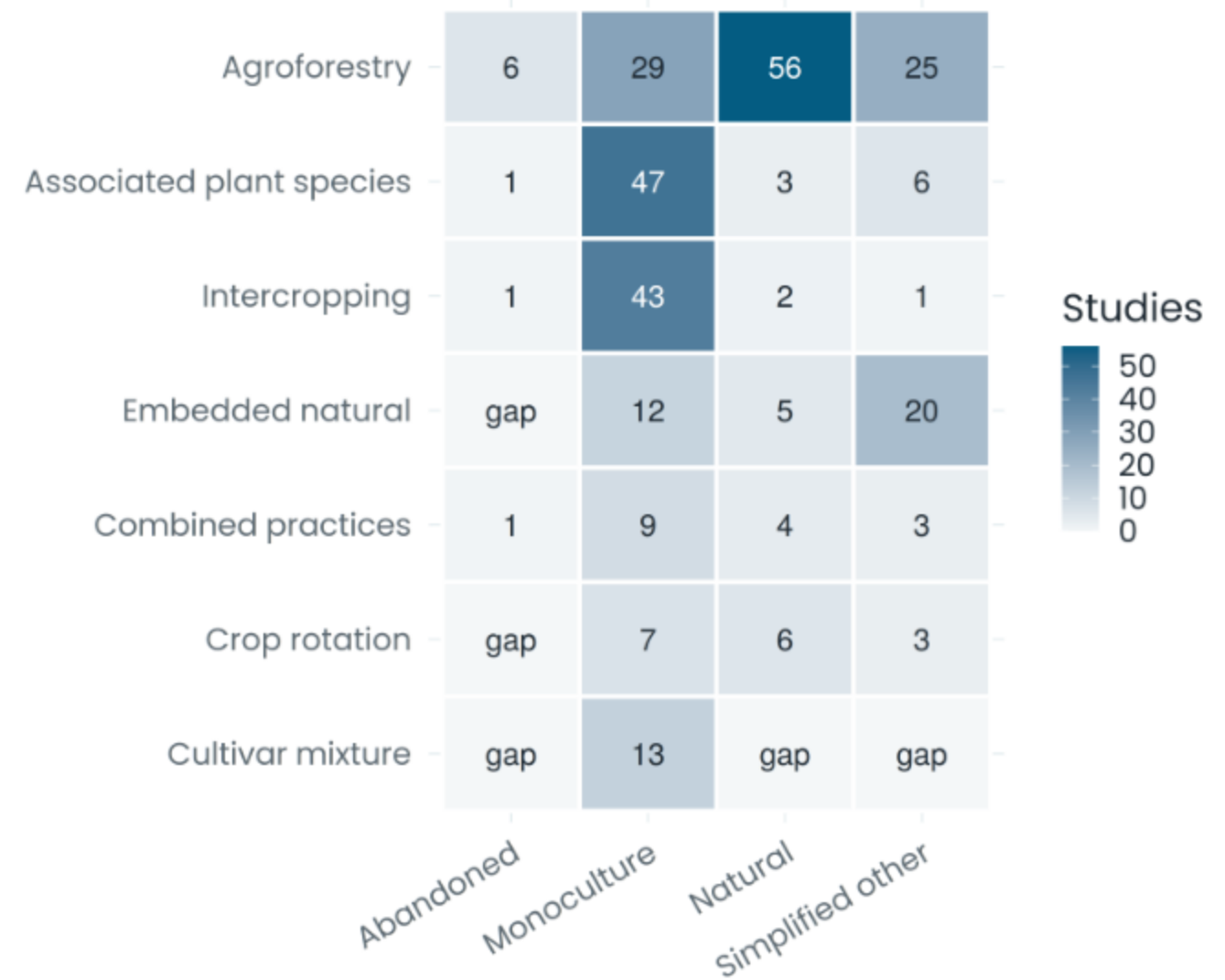
One study can report dozens of comparisons (taxa, sites, dates). **Rows are not studies — and they are not independent.** Count with `n_distinct(study_id)`.

Same data, two maps

WRONG: rows, red-white-blue, gaps just missing



BETTER: studies, sequential colours, gaps shown



What an evidence map tells you — and what it does not

It tells you

- **Where** evidence accumulates (clusters)
- **Where** it is missing (gaps)
- **How** it was produced (designs, metrics, validity)
- Whether a **meta-analysis** is feasible, and on what

It does not tell you

- **Whether** a practice works, or how much
- Whether results are **consistent**
- Whether a gap is a **priority** (it may be agronomically irrelevant)

Many studies $\neq$ strong effect. Do not “vote count”.

Your 30 minutes

Time	Task	Format
14:08	Open the pack, load the data, count studies	Alone
14:13	Exercise 1 — first evidence map (fill the ___)	Alone
14:25	Exercise 2 — add a third dimension: <i>organisms</i> or <i>validity</i>	Pairs
14:35	2–3 pairs show their map	All
Bonus	Where (world map) and when (cumulative curves)	Later

Open `meta_analysis_course.Rproj` (the course kit), then `td_evidence_map.R`. Instructions and folded solutions: **the practical web page**.

KEY MESSAGES

- 1 | **Count studies, not rows** — rows are non-independent comparisons.
- 2 | **Show the zeros:** in an evidence map, the gap *is* the result.
- 3 | **Sequential colours and numbers in tiles:** readable by everyone.
- 4 | **A map describes the evidence; it does not say what works.** Your codebook sets its axes.

In the online notebook

Each point of this session developed further, with the same data, more pitfalls and exercises with solutions:

- The running example and first data checks
- Evidence maps: heat maps, gaps and a third dimension
- Temporal analyses
- Spatial analyses

literaturesynthesis.github.io/notebook

References

Collaboration for Environmental Evidence. 2022. [Guidelines and standards for evidence synthesis in environmental management. Version 5.1.](#) (A. S. Pullin, G. K. Frampton, B. Livoreil, and G. Petrokofsky, Eds.). Collaboration for Environmental Evidence.

James, K. L., N. P. Randall, and N. R. Haddaway. 2016. [A methodology for systematic mapping in environmental sciences.](#) Environmental Evidence 5:7.

Jones, S. K., A. C. Sánchez, S. D. Juventia, and N. Estrada-Carmona. 2021. [A global database of diversified farming effects on biodiversity and yield.](#) Scientific Data 8:212.